جامعة مصر للمعلوماتية
EGYPT UNIVERSITY OF INFORMATICS

FACULTY OF COMPUTING & INFORMATION SCIENCES

C-SW312: Introduction to Software Engineering
Fall 2026

General Guidelines for Project Deliverables, Submission, and Discussion

The following guidelines apply to **ALL project deliverables** throughout the course. Students are expected to carefully read and strictly adhere to these instructions.

General Requirements

1. Students are expected to demonstrate **creativity, critical thinking, and sound technical reasoning** when developing their solutions and models. All diagrams should be sufficiently detailed, accurate, consistent, and self-explanatory.
2. Students must make reasonable and well-justified assumptions whenever the provided information is insufficient. **All assumptions must be explicitly documented** in the Technical Report and, where appropriate, included as annotations or comments in the corresponding diagrams.
3. All models and diagrams must be developed using appropriate professional modeling tools, preferably UML modeling tools where applicable, and must follow the relevant modeling standards and notation.
4. All submitted diagrams must be clear, readable, properly labeled, and consistent with the corresponding descriptions and other models presented in the report.

Deliverable Submission Requirements

5. For **EACH deliverable**, every group must submit a comprehensive, well-structured **Technical Report** containing all required steps, explanations, tables, models, and diagrams specified in the respective deliverable description.
6. The Technical Report must clearly explain the proposed solutions, justify important design and modeling decisions, and explicitly document all relevant assumptions.
7. The Technical Report must be submitted in **PDF format**. Handwritten reports, diagrams, or models will **NOT** be accepted.
8. In addition to the Technical Report, students must submit the **original editable project files** of all diagrams and models developed using the selected modeling tool. These files should preferably be submitted as a single compressed (ZIP) folder.
9. All submissions must be uploaded exclusively through **Canvas**, using the designated submission link created for each deliverable.

Academic Integrity, Plagiarism, and AI Usage Policy

10. **The use of ANY Artificial Intelligence (AI) tool is STRICTLY PROHIBITED** in preparing, generating, modifying, or completing any part of the project deliverables. This includes, but is not limited to, Generative AI tools, Large Language Models (LLMs), AI coding assistants, AI-powered writing assistants, and AI-based diagram or model generation tools.
11. All submitted reports, diagrams, models, explanations, and other deliverable components must represent the **original work and intellectual effort of the students**. Students are expected to independently develop their solutions and demonstrate a thorough understanding of every aspect of their submissions.
12. **All submitted work may be screened using plagiarism detection and AI-generated content detection tools.** Suspected violations may also be investigated through manual review, examination of project files, and oral discussions with group members.
13. **Plagiarism, unauthorized AI usage, and other forms of academic misconduct will be treated with the utmost seriousness.** Any confirmed violation will result in **ZERO marks for the respective deliverable** and referral for **disciplinary action at the Faculty level**, in accordance with the applicable academic regulations.
14. Every group member is expected to understand and be able to explain the submitted work. Inability to justify modeling decisions, explain diagrams, or demonstrate an understanding of the submitted solutions may trigger further academic review.

Submission Deadlines and Late Submission Policy

15. **NO LATE SUBMISSIONS WILL BE ACCEPTED.** Any deliverable submitted after the announced deadline will automatically receive **ZERO marks**, subject to any formally approved exceptions required by University or Faculty regulations.
16. Students are strongly advised **NOT to wait until the last minute** to submit their work. Internet connectivity issues, technical difficulties, upload failures, or other last-minute problems will **NOT** normally be accepted as valid excuses for missing the deadline.
17. Students are responsible for ensuring that all required files are correctly uploaded, accessible, and complete before the submission deadline.

Deliverable Discussions and Feedback

18. **Deliverable discussions will be scheduled after the submission deadline.** Students will be expected to present, explain, and justify their submitted solutions, models, and design decisions.
19. **All group members must actively participate in the discussions** and demonstrate a clear understanding of the work submitted by their group.

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Guidelines for Deliverable #2: User Stories and Use Cases

Relevant Chapter: Chapter 3 – User Stories and Use Cases
Submission Deadline: Thursday, 5 November 2026, at 11:59 PM

Objective

The objective of this deliverable is to apply requirements elicitation and analysis techniques to the selected case study by identifying stakeholders' needs, formulating user stories and acceptance criteria, and systematically identifying and modeling system use cases using the **Event Decomposition Technique**.

Students are expected to demonstrate their ability to translate stakeholder needs and business events into clearly defined functional requirements.

Preliminary Activity: Stakeholder Role-Playing

Divide your project team into two subgroups:

- **Group #1 – System Analysts:** Responsible for eliciting, analyzing, clarifying, and documenting system requirements.
- **Group #2 – Users/Stakeholders:** Responsible for representing the different stakeholder categories identified in Deliverable #1, explaining their needs, describing their work-related activities, and communicating their expectations of the proposed system.

Conduct a simulated requirements elicitation session in which the system analysts interview the users/stakeholders to understand their needs, goals, and expected interactions with the system.

Expected Output: A brief summary of the requirements elicitation session, including the stakeholder roles represented, key questions raised, and the requirements identified.

Part I: User Stories

Based on your selected case study and the **four stakeholder categories identified in Deliverable #1**, perform the following activities:

1. Identify and Formulate User Stories

For each stakeholder category, identify relevant user roles and construct user stories describing the work-related tasks that users need to perform to achieve specific goals or outcomes.

Each user story must follow the standard template:

"As a [role], I want to [goal], so that [benefit]."

User stories should focus on the stakeholder's needs and intended outcomes rather than technical implementation details.

Minimum Requirement: Develop at least **eight (8) user stories**, with at least **two (2) user stories associated with each of the four stakeholder categories** identified in Deliverable #1.

Where a stakeholder category does not directly interact with the system, clearly explain how its needs are represented by an appropriate user role or system requirement.

2. Define Acceptance Criteria

For **EACH user story**, identify and document clear, specific, and testable **Acceptance Criteria** that determine when the user story can be considered successfully implemented.

Acceptance criteria must follow the **Given–When–Then** format:

- **Given:** The initial conditions or context.
- **When:** The action or event that occurs.
- **Then:** The expected system behavior or outcome.

Each user story should include sufficient acceptance criteria to cover its main expected behavior and relevant alternative or exceptional conditions.

3. Evaluate the Quality of User Stories

For **EACH user story**, evaluate its quality and effectiveness using the **INVEST criteria**:

- **I – Independent:** Can the story be developed with minimal dependencies on other stories?
- **N – Negotiable:** Does the story describe a need without unnecessarily prescribing its implementation?
- **V – Valuable:** Does the story provide meaningful value to a user or stakeholder?
- **E – Estimable:** Is the story sufficiently clear to estimate the effort required?
- **S – Small:** Is the story appropriately scoped and manageable?
- **T – Testable:** Can its successful implementation be verified through acceptance criteria?

Document your evaluation and briefly justify any necessary refinements to improve the quality of the user stories.

Part II: Use Case Identification – Event Decomposition Technique

Apply the **Event Decomposition Technique** systematically to identify the functional requirements of your proposed system.

You are required to demonstrate the application of each step, using the relevant checklists introduced in Chapter 3.

1. Identify External Events

Identify the **external events** occurring in the system environment that require a response from the proposed system.

For each identified external event:

- Describe the event.
- Identify the external actor or stakeholder responsible for triggering it.
- Identify and name the corresponding use case that represents the system's response.

Use the **External Events Checklist** provided in the lecture slides to ensure that relevant events have been considered.

2. Identify Temporal Events

Identify the **temporal events** that require the system to perform certain activities at predetermined times or intervals.

For each identified temporal event:

- Describe the event.
- Specify the time, date, frequency, or scheduling condition that triggers it.
- Identify and name the corresponding use case.

Use the **Temporal Events Checklist** provided in the lecture slides.

3. Identify State Events

Identify the **state events** that occur when the system must respond to a significant change in the condition or status of an entity.

For each identified state event:

- Describe the event.
- Identify the entity whose state changes.
- Specify the state change or condition that triggers the system response.
- Identify and name the corresponding use case.

4. Apply the Perfect Technology Assumption

Review all identified events and use cases to determine whether they represent genuine business requirements.

Apply the **Perfect Technology Assumption** to avoid including use cases that primarily address technological or system-control concerns rather than business activities.

For this deliverable, do not include use cases such as:

- Login or logout
- Change password
- Backup or restore database

These technical and security-related functions will be addressed at a later stage.

5. Construct the Event–Use Case Table

Prepare a consolidated table documenting all identified events and their corresponding use cases.

The table must include the following columns:

Event ID	Event Description	Event Type	Triggering Actor / Condition	Corresponding Use Case
E01	...	External	...	...
E02	...	Temporal	...	...
E03	...	State	...	...

Ensure that all identified events are appropriately categorized as **External, Temporal, or State events**.

6. Develop Brief Use Case Descriptions

For **EACH identified use case**, prepare a **Brief Use Case Description** using the format introduced in Chapter 3.

Each description should include, at minimum:

- **Use Case ID**
- **Use Case Name**
- **Brief Description:** A concise explanation of the system activity and its intended outcome.
- **Actor(s):** The external actor(s) interacting with the system, where applicable.
- **Triggering Event:** The event that initiates the use case.

Use case names should clearly describe the system functionality, preferably using a **verb–noun format**, such as *Place Order*, *Generate Report*, or *Update Customer Information*.

7. Develop the UML Use Case Diagram

Graphically model the identified use cases using a professional **UML modeling tool**.

Your UML Use Case Diagram must:

- Clearly identify the **system boundary**.
- Include all relevant **actors** interacting with the system.
- Represent the identified **use cases** within the system boundary.
- Correctly represent the **associations between actors and use cases**.
- Apply appropriate UML notation and naming conventions.
- Be consistent with the Event–Use Case Table and Brief Use Case Descriptions.

Important: Not every stakeholder is necessarily an actor. Only stakeholders or external systems that directly interact with the proposed system should be represented as actors.

Part III: Project Planning and Collaboration

1. Jira Project Timeline

Prepare or update your **Jira project plan** to reflect the activities required to complete Deliverable #2.

Your Jira project should include:

- Tasks corresponding to the main deliverable activities.
- Assignment of responsibilities among group members.
- Planned start dates and deadlines.
- Task status and progress tracking.

Include appropriate evidence of your Jira timeline and task assignments in the Technical Report.

2. GitHub Collaboration

Use **GitHub** to organize and share the deliverable-related project files among group members. Ensure that the repository contains the relevant project artifacts and reflects meaningful contributions from group members.

Provide the GitHub repository link in your Technical Report.

Expected Outcomes

Upon completion of Deliverable #2, each group is expected to produce:

1. A summary of the stakeholder role-playing and requirements elicitation activity.

2. At least **eight user stories**, covering the four stakeholder categories identified in Deliverable #1.
3. Acceptance criteria for each user story using the **Given–When–Then** format.
4. An evaluation of each user story against the **INVEST criteria**.
5. A systematic application of the **Event Decomposition Technique**, covering external, temporal, and state events.
6. A consolidated **Event–Use Case Table**.
7. **Brief Use Case Descriptions** for all identified use cases.
8. A comprehensive **UML Use Case Diagram**.
9. An updated **Jira project timeline** and evidence of GitHub collaboration.

Deliverable Submission Format

Each group must submit the following through Canvas:

A. Technical Report (PDF)

A well-organized report containing all activities, tables, descriptions, diagrams, assumptions, and justifications specified above.

B. UML Project Files

The original editable files containing the developed UML Use Case Diagram, preferably submitted in a compressed ZIP folder.

C. Project Management and Collaboration Evidence

The Jira timeline and GitHub repository link, included in the Technical Report.

Important Notes

- All requirements outlined in the **General Guidelines for Project Deliverables, Submission, and Discussion** apply to this deliverable.
- The use of AI tools is strictly prohibited, in accordance with the course's academic integrity policy.
- All group members must understand and be prepared to explain and justify the submitted work during the deliverable discussion.
- **Late submissions will not be accepted.**